Forensic Science International 236 (2014) 30–37

Contents lists available at ScienceDirect

Forensic Science International

journal homepage: www.elsevier.com/locate/forsciint

Evaluation of fingermark detection sequences on paper substrates

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ARTICLE INFO

Article history:
Received 22 August 2013
Received in revised form 18 December 2013
Accepted 19 December 2013
Available online 6 January 2014

Keywords:
Fingerprints
Porous surfaces
1,2-Indanedione
Ninhydrin
1,8-Diazafluoren-9-one
Physical developer
Nile red

ABSTRACT

It is generally accepted that the amino acid reagent consisting of 1,2-indanedione and a catalytic amount of zinc chloride, referred to as IND-Zn, is the single best method for the detection of latent fingermarks on paper substrates and that ninhydrin is of limited value when used in sequence after this reagent. However, recent research has suggested that the sequence 1,8-diazafluoren-9-one (DFO) followed by ninhydrin may actually produce a greater number of fingermarks than IND-Zn on its own or IND-Zn followed by ninhydrin.

This study focussed on the evaluation of two fingermark detection sequences for porous surfaces: (1) IND-Zn followed by ninhydrin, physical developer (PD) and the lipid stain nile red; and (2) DFO followed by ninhydrin, PD and nile red. The evaluation was undertaken using a range of latent fingermark donors and on a number of paper substrates that are commonly encountered in Australia. In addition, a pseudo-operational trial was completed on 5-year-old university examination booklets. Parallel studies were undertaken at two locations: Sydney (temperate, coastal climate) and Canberra (relatively dry, continental climate).

The results of the donor study indicated that there was a negligible difference in performance between the two sequences across all paper types and all time periods evaluated. When considering individual reagents, IND-Zn generally developed better quality fingermarks compared to DFO; however, ninhydrin had a greater enhancement effect on DFO developed marks than after IND-Zn. In the pseudo-operational trials, the IND-Zn sequence outperformed the DFO sequence. Nile red did not develop any additional marks at the end of each sequence and, as a result, the use of this technique at the end of a full sequence is of questionable value.

The overall outcome was that the sequence IND-Zn followed by ninhydrin and PD is recommended for the processing of common paper substrates under the conditions typically experienced at the two locations studied.

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1. Introduction

The detection and enhancement of latent fingermarks requires the application of an appropriate sequence of methods that will depend on the nature of the surface, the circumstances of the case under investigation, and the resources available to the fingerprint technician. Such sequences, consisting of complementary detection methods from least destructive to more destructive, need to be optimised and validated under local conditions before casework implementation. For paper substrates, the generally accepted approach is to apply non-destructive optical methods first, followed by one or more amino acid reagents, then a method – such as physical developer – that targets any sebaceous material that may be present [1].

The traditional amino acid reagent for fingermark detection is ninhydrin, which was first proposed for this application in 1954 [2]. Ninhydrin remains the most common chemical method for the treatment of paper substrates despite significant research efforts directed at the synthesis of ninhydrin analogues [3,4]. The first potential alternative to ninhydrin to be introduced into routine casework was 1,8-diazafluoren-9-one (DFO), originally proposed as a fingermark reagent in 1990 [5]. However, rather than replacing ninhydrin, DFO proved to be effective when used in a sequence prior to ninhydrin treatment, with early indications that ninhydrin can develop additional fingermarks not detected by DFO [6].

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http://dx.doi.org/10.1016/j.forsciint.2013.12.028

More recently, in 1997, 1,2-indanedione (IND) was introduced as a novel amino acid reagent that showed significant potential [7]. Various research groups around the world investigated the application of this reagent for fingermark detection on paper substrates, with some groups reporting superior results compared to DFO [8,9], while others found that DFO was more sensitive [10]. A breakthrough occurred when it was determined that the addition of a catalytic amount of zinc chloride to the IND working solution resulted in significantly improved results due to the preferential formation of the desired, highly luminescent reaction product [11,12]. This new formulation, referred to as IND-Zn, was subsequently adopted for routine use by a number of agencies including the Australian Federal Police [13]. As for DFO, IND-Zn can be used in sequence prior to conventional ninhydrin treatment; however, anecdotal evidence has indicated that the results achieved with IND-Zn are rarely improved by subsequent treatment with ninhydrin.

A recent study by Porpiglia et al. looked at the effectiveness of the ninhydrin analogue 5-methylthioninhydrin as a reagent for fingermark detection on paper substrates under UK conditions [14]. As part of this study, the authors compared a number of sequences, including IND-Zn→ninhydrin and DFO→ninhydrin, for their ability to develop fingermarks collected from 29 participants on six selected paper types. The indication was that IND-Zn was the most effective reagent of those studied if a single treatment is employed. However, the sequence DFO→ninhydrin developed 87% of the test fingermarks while the sequence IND-Zn→ninhydrin only developed 78%. The authors concluded that the sequence DFO→ninhydrin was more effective than the alternatives investigated.

Physical developer (PD), the preferred reagent for targeting sebaceous material in latent fingermarks on paper, was developed in the 1970s by the Atomic Weapons Research Establishment under contract to the UK Police Scientific Development Branch [15,16]. PD is used at the end of the detection sequence and it can develop fingermarks that remain undetected using amino acid reagents [1,17]. However, the technique is costly and time-consuming, and requires significant expertise to achieve optimum results. Single metal deposition (SMD) has also been proposed for use as a fingermark detection method on porous substrates [18]. While SMD is easier to apply than multimetal deposition (MMD), it remains a labour-intensive technique that has not been widely adopted at this point in time.

A number of research groups have investigated lipid stains such as Oil Red O (ORO) as a simpler alternative to processes such as PD or SMD [19,20]. While it has been suggested that ORO can be used in sequence prior to PD treatment, there is an adverse effect on the PD results due to higher background development [20]. A recent study by Braasch et al. indicated that the luminescent lipid stain nile red can be used in sequence after PD and may develop additional fingermarks [21]. However, further research is required before nile red can be considered for use in casework at the end of a full detection sequence on paper (i.e., including amino acid reagents as opposed to PD only).

The aim of this study was to evaluate two fingermark detection sequences, one incorporating IND-Zn and the other incorporating DFO (Fig. 1), on common Australian paper substrates and under Australian conditions. This was achieved by applying these sequences in donor trials, on fingermarks deposited by a number of individuals on 4 different paper substrates, and in pseudo-operational trials on pages taken from 5-year-old university examination booklets. The use of nile red was preferred over ORO as it can be used in sequence after PD, thus not impacting on the performance of PD, and it produces luminescent fingermarks rather than a simple coloration. The study was conducted in parallel at two different geographic locations (Canberra and Sydney) to determine if there were variations in the performance of each sequence due to climatic conditions.

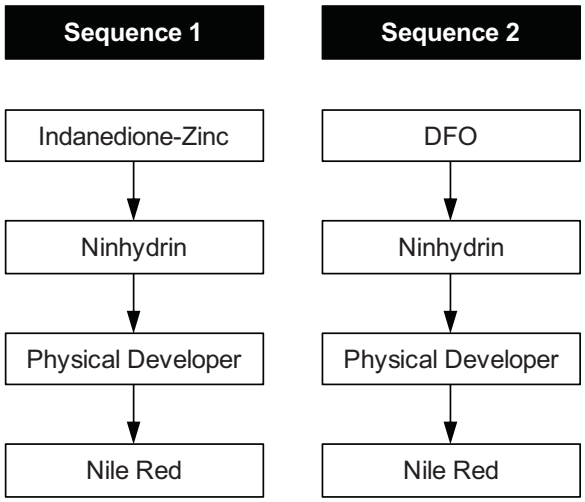


Fig. 1. The two sequences considered in this study for their ability to detect latent fingermarks on a range of paper substrates.

2. Materials and methods

2.1. Chemicals

IND, DFO, ninhydrin, n-dodecylamine acetate and Synperonic N were purchased from Optimum Technology (Canberra, Australia). Silver nitrate, citric acid, maleic acid, ammonium iron(II) sulphate hexahydrate, iron(III) nitrate nonahydrate and nile red were purchased from Sigma–Aldrich Pty. Ltd. (Sydney, Australia). Ethyl acetate, acetic acid, ethanol, methanol, and zinc chloride were purchased from Chem-Supply Pty. Ltd. (Gillman, Australia). HFE-7100 and HFC-4310mee were purchased from Novaline (Taren Point, Australia). For the Canberra study, high-purity water was obtained from a Satorius arium 611 water purification system. In Sydney, high-purity water was obtained from a high-throughput three-stage water filtration system (AKF 300 activated carbon filter; Bewades 58 LC UV disinfection system; Vertex SS-360HR reverse osmosis water purification system).

2.2. Fingermark detection methods

For IND-Zn, DFO, ninhydrin and PD, the reagent formulations and development procedures currently in use by the Australian Federal Police were employed [13]. IND-Zn treated samples were processed in a dry heat press at 160 °C (±5 °C) for 10 s. DFO treated samples were processed in a dry heat press at 180 °C (±5 °C) for 10 s. For ninhydrin treated samples, development was allowed to proceed at room temperature over 24–48 h. The PD process involved three initial deionised water washes, treatment in a maleic acid solution, a further water wash, PD development until optimum fingermark contrast was observed, and then three final deionised water washes. Nile red treatment was performed using the modified working solution and general development procedure described by Braasch et al. [21]. Developed fingermarks were visualised and digitally recorded under the conditions indicated in Table 1, with a Polilight PL-500 (Rofin Pty. Ltd., Australia) used as the light source.

2.3. Phase 1: donor trials

For the donor trial, four common paper substrates were chosen. These were: (A) white virgin (non-recycled) printer/copier paper (Reflex, Australia); (B) 100% recycled white printer/copier paper

Table 1
Observation conditions used for the recording of developed fingermarks.

Development method	Polilight PL-500 band	Band-pass barrier filter
IND-Zn	530 nm	590 nm
DFO	530 nm	590 nm
Ninhydrin	White light	565 nm
PD	White light	–
Nile Red	490 nm	550 nm

(Reflex, Australia); (C) lined notepad paper (Tudor, Australia); and (D) Kraft envelope paper (Officeworks, Australia). A total of 12 fingermark donors (6 males, 6 females) were employed in the Canberra study, while 6 fingermark donors (3 males, 3 females) were employed in the Sydney study (Table 2). Latent fingermarks, collected from these donors on the four paper substrates and stored in closed envelopes in the dark, were aged for 2 weeks, 1 month and 3 months prior to treatment.

Fingermarks were collected from each donor as controlled 4-step depletion sets, with two left-hand/right-hand sample sets collected for each substrate and each age bracket (Fig. 2), with a minimum of a 30-min time gap between sets to ensure regeneration of natural secretions on the hands of the donor. Only natural fingermarks were collected (i.e., no artificial charging of the fingers with eccrine or sebaceous secretions), with the donor simply asked to rub their hands together to homogenise the secretions present prior to the collection of each sample set. Each paper strip, consisting of a depletion series of 3-finger impressions, was aged for the required time period then cut down the middle, with one side subjected to the IND-Zn detection sequence (Sequence 1) and the other side subjected to the DFO detection sequence (Sequence 2; Fig. 1).

After each step in each sequence, the two corresponding half-strips were reassembled and imaged under the conditions indicated in Table 1. The quality of the fingermark development was then assessed for each 3-finger impression using the comparative University of Canberra (UC) scoring system described by McLaren et al. [22] (Table 3). Each half-impression was also assessed using the absolute Home Office (HO) scale described by Sears et al. [23] (Table 4). Each comparison therefore generated one comparative rating (UC scale) and two absolute ratings (HO scale), one for the Sequence 1 half-impression and one for the Sequence 2 half-impression. Two assessment scales were applied as one scale on its own cannot capture all of the required information. The HO scale assesses the overall quality of a fingermark but it is not good at comparing two methods when the differences are small. The UC scale assesses differences in performance, even relatively minor ones, but does not provide an overall assessment of fingermark quality.

Table 2
Fingermark donors used in the Canberra and Sydney trials.

Study Location	Donor number	Gender	Age	Occupation
Canberra	1	Male	22	Sport coordinator/student
	2	Male	22	Student
	3	Male	21	Client services officer/student
	4	Male	22	Swimming instructor/student
	5	Female	21	Pharmacy dispensary technician/student
	6	Male	22	Swimming Instructor/student
	7	Female	22	Waitress/student
	8	Female	22	Student
	9	Female	18	Student
	10	Female	22	Receptionist/student
	11	Female	24	Librarian/student
	12	Male	23	Program support Officer/student
Sydney	1	Female	25	Student
	2	Male	21	Student
	3	Female	27	Academic
	4	Male	27	Software engineer
	5	Female	22	Student/café worker
	6	Male	22	Student

Table 3
Comparative UC scale [22] used for the assessment of the results from the donor trials.

Rating	Definition
−2	Half-impression from the DFO sequence exhibits far greater ridge detail and/or contrast than the corresponding half-impression from the IND-Zn sequence
−1	Half-impression from the DFO sequence exhibits slightly greater ridge detail and/or contrast than the corresponding half-impression from the IND-Zn sequence
0	No significant difference between the corresponding half-impressions
+1	Half-impression from the IND-Zn sequence exhibits slightly greater ridge detail and/or contrast than the corresponding half-impression from the DFO sequence
+2	Half-impression from the IND-Zn sequence exhibits far greater ridge detail and/or contrast than the corresponding half-impression from the DFO sequence

Table 4
Absolute HO scale [23] used for the assessment of the results from the donor trials. (Note that each half-impression was assessed in isolation against this scale.).

Rating	Definition
0	No evidence of a fingermark
1	Weak development; evidence of contact but no ridge detail
2	Limited development; about 1/3 of ridge detail present but probably cannot be used for identification purposes
3	Strong development; between 1/3 and 2/3 of ridge detail present; identifiable fingermark
4	Very strong development; full ridge detail present; identifiable fingermark

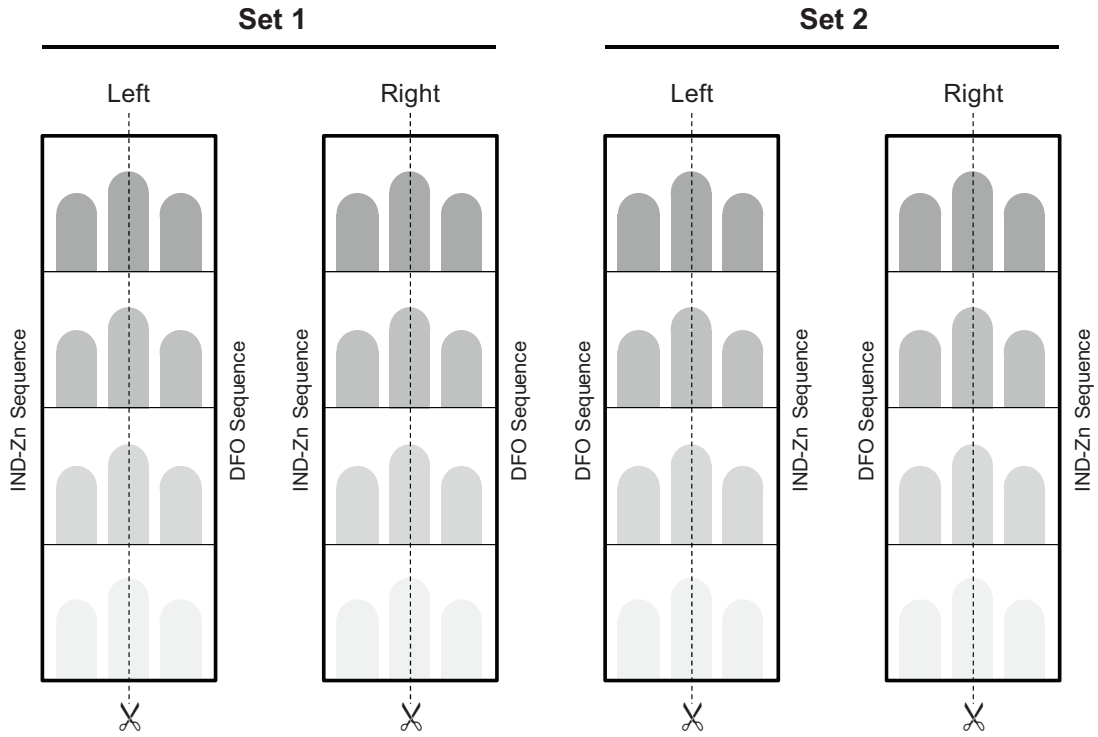


Fig. 2. Fingermark sample sets (as 4-step depletions) collected from each donor for each paper substrate and each time period. There was a minimum of a 30-min time gap between sets to ensure regeneration of natural secretions on the hands of the donor.

A subset of the assessment results was double-checked by a fingerprint identification expert (co-author BC) to ensure consistency and adherence to the scale definitions.

2.4. Phase 2: pseudo-operational trials

For the pseudo-operational trials, de-identified 5-year-old examination booklets from the University of Canberra (UC) and the University of Technology, Sydney (UTS) were used as test samples. It was expected that the white lined pages in these booklets had been touched by both the students completing the examinations and the staff members responsible for marking the examination responses.

Table 5
Recorded temperature and relative humidity within the Canberra and Sydney laboratories over the course of the study.

Location	Temperature (°C)		Relative humidity (%)	
	Range	Average	Range	Average
Canberra (UC)	19–24	22	43–64	50
Sydney (UTS)	17–25	22	45–76	61

Table 6
Average results across all 4 paper substrates for the Canberra donor trial. (*Note*: Nile red has been excluded given that no results were obtained with this technique.).

Age	Reagent	Avg. UC score	Avg. HO score	
			Sequence 1	Sequence 2
2 weeks	IND-Zn or DFO	0.55	2.60	2.16
	Ninhydrin	−0.24	1.08	1.30
	PD	−0.05	0.06	0.11
	Overall sequence	0.08	1.24	1.19
1 month	IND-Zn or DFO	0.64	2.74	2.33
	Ninhydrin	−0.08	1.56	1.62
	PD	−0.05	0.09	0.13
	Overall sequence	0.17	1.46	1.36
3 months	IND-Zn or DFO	0.30	2.88	2.60
	Ninhydrin	−0.28	1.58	1.83
	PD	−0.07	0.37	0.41
	Overall sequence	−0.01	1.61	1.62
All ages combined	IND-Zn or DFO	0.50	2.74	2.36
	Ninhydrin	−0.20	1.41	1.58
	PD	−0.06	0.17	0.22
	Overall sequence	0.08	1.44	1.39

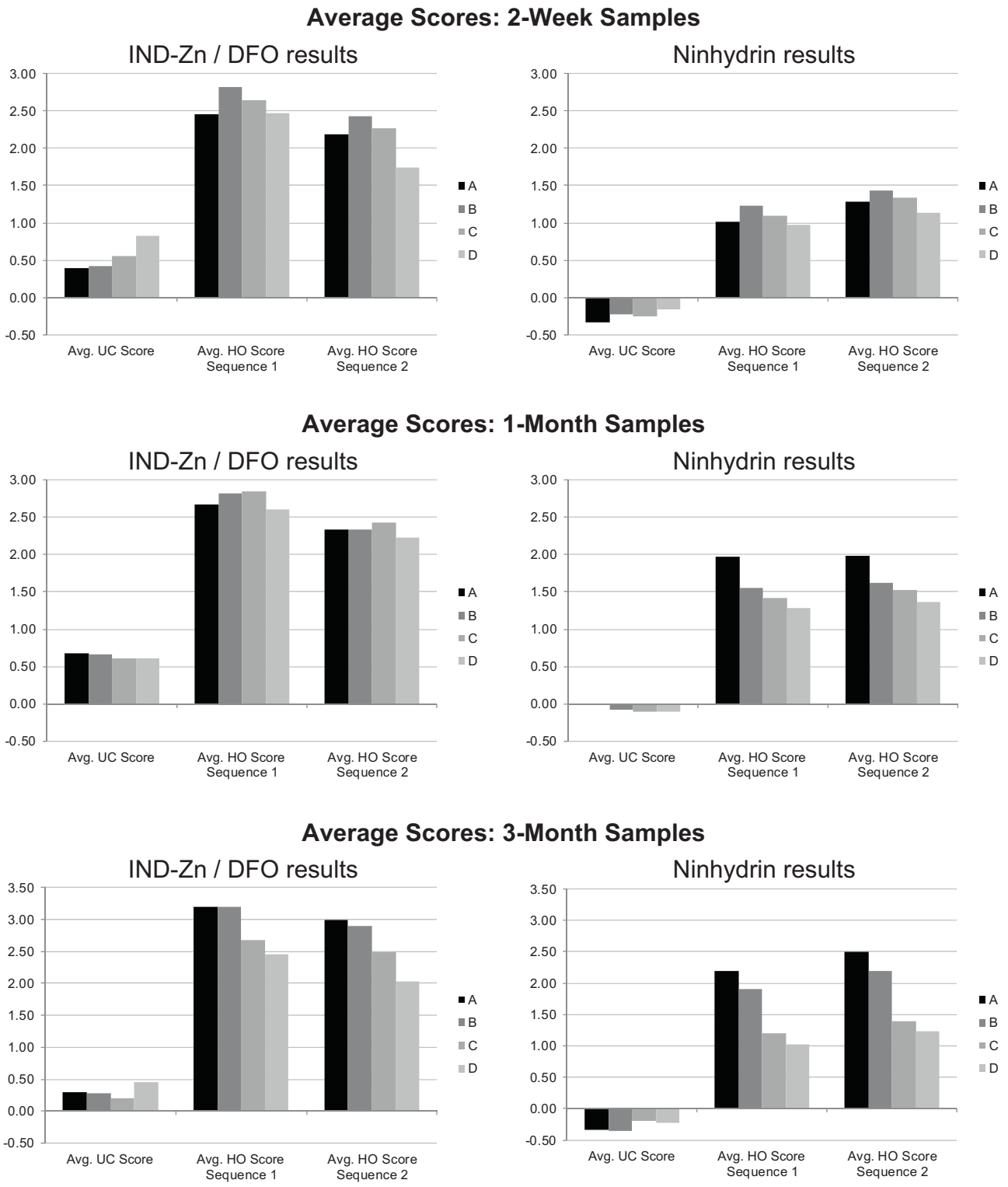


Fig. 3. Average UC and HO scores recorded in the Canberra donor trial after IND-Zn or DFO and after subsequent treatment with ninhydrin. Results are displayed for each paper substrate (A = white virgin copy paper; B = white recycled copy paper; C = lined notepad paper; D = Kraft envelope paper).

For the Canberra trial, 20 booklets were selected at random, with 4 pages taken from each; two pages per booklet were subjected to the IND-Zn sequence while the other two pages were subjected to the DFO sequence (resulting in 40 pages or 80 sides per sequence). For the Sydney trial, 10 booklets were selected at random, with 2 pages taken from each; one page per booklet was subjected to the IND-Zn sequence while the other page was subjected to the DFO sequence (resulting in 10 pages or 20 sides per sequence). After the application of each detection method, the pages were examined under the conditions indicated in Table 1. The number of developed fingermarks of HO value 2 or above (Table 4) and not developed by any previous method applied in the

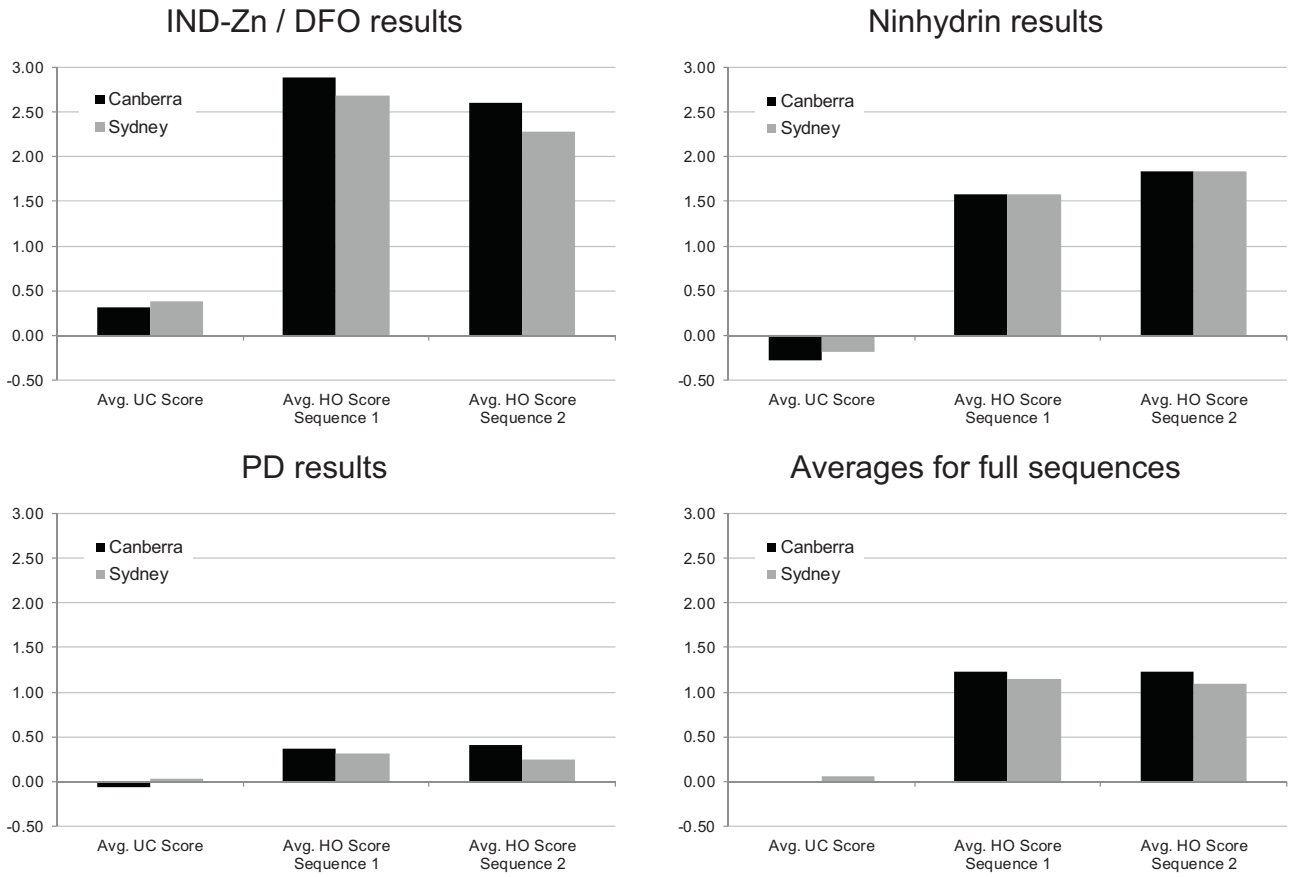


Fig. 4. Comparison of Canberra and Sydney average UC and HO scores for the 3-month samples in the donor trials (all 4 paper substrates combined). (*Note*: Nile red has been excluded given that no results were obtained with this technique.).

sequence was then tabulated. In this manner, it was possible to determine the number of additional fingermarks developed by each method in the sequence.

2.5. Environmental conditions

Laboratory temperature and humidity data were continuously recorded at the two locations (Sydney and Canberra) over the course of both the Phase 1 and Phase 2 trials.

3. Results and discussion

3.1. Environmental conditions

Over the period of the trials, the laboratory conditions at the two locations (for the storage and treatment of the fingermark samples) were similar, with the only significant difference being a slightly higher average relative humidity in the Sydney facility (Table 5), being 61% compared to 50% in Canberra.

3.2. Phase 1: donor trials

The amino acid reagents IND-Zn, DFO and ninhydrin performed well across all 4 substrates, resulting in good fingermark development for most donors and depletion sets. PD, however, exhibited an adverse reaction with three of the paper substrates (the exception being the notepad paper), with only a limited number of fingermarks being developed from a small number of donors. Nile red, used at the end of both sequences, failed to provide any fingermark development. This suggests that the sebaceous fraction targeted by this lipid stain had been removed earlier in the detection sequences, possibly due to the solvents used in the amino acid reagents and/or the heat treatment required for the IND-Zn and DFO development. A short experiment confirmed that the prolonged dipping of latent fingermarks in the carrier solvents employed for IND-Zn, DFO and ninhydrin had a detrimental effect on subsequent nile red development. This need to be further investigated to determine if such effects can be mitigated.

The lack of results obtained with nile red, both in the Canberra and Sydney trials, meant that only zero comparative and absolute scores were achieved for this technique. As such, nile red has been excluded from the presentation of results below.

Average UC and HO scores were calculated for each paper type at each time period, after the application of each step in the two sequences. For the Canberra study, these results are summarised in Table 6 (all substrates combined) and in Fig. 3 (average values after IND-Zn/DFO and ninhydrin). A comparison of the Canberra and Sydney results is provided in Fig. 4 for the 3-month samples (all substrates combined). The general trends discussed below were observed in both Canberra and Sydney.

Regardless of the paper substrate, IND-Zn gave average UC scores that were positive, indicating generally better results when compared with the associated DFO treated half-impressions. This observation was further supported by average HO scores for IND-Zn that were higher than the average HO scores for DFO. This suggests that, for the substrates and fingermarks tested, IND-Zn developed better quality fingermarks than those developed with DFO. Ninhydrin, however, demonstrated better performance when used after DFO than after IND-Zn. This is indicated by negative UC scores and average HO scores that were higher for the Sequence 2

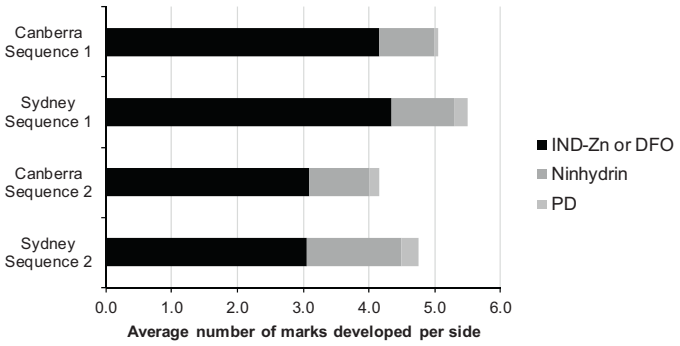


Fig. 5. Results from the pseudo-operational trials on 5-year-old university examination booklets. (To be counted, developed marks had to have a HO rating of 2 or above.)

half-impressions, regardless of the paper substrate tested and regardless of fingermark age (Figs. 3 and 4). The application of PD resulted in essentially zero average UC scores, indicating no significant difference between corresponding half-impressions. However, given the limited results achieved with PD, very low average HO scores (less than 0.5) were obtained. Comparing the two sequences overall, average UC scores close to zero and similar average HO scores indicated that neither sequence – as a whole – had outperformed the other (Fig. 4).

3.3. Phase 2: pseudo-operational trials

The results from the pseudo-operational trials, conducted on pages taken from 5-year-old university examination booklets, are summarised in Table 7 and Fig. 5. In the Canberra study, IND-Zn developed 34% more fingermarks than DFO. While in the Sydney study, IND-Zn developed 43% more fingermarks than DFO. However, as observed with the donor trials, ninhydrin proved to be more effective when used after DFO than when used in sequence after IND-Zn. In Canberra, ninhydrin developed an additional 20% more marks after IND-Zn, and an additional 29% more marks after DFO. In Sydney, ninhydrin developed an additional 22% more marks after IND-Zn, and an additional 48% more marks after DFO. PD contributed only a small number of additional fingermarks in both the Canberra and Sydney studies. As for the donor trials, nile red used at the end of each sequence failed to develop any fingermarks on the pages taken from the examination booklets.

IND-Zn as an individual method developed – on average – a similar number of fingermarks as the sequence DFO followed by ninhydrin. As a whole sequence, Sequence 1 outperformed Sequence 2, developing a higher number of fingermarks per side of each treated page in both the Canberra and Sydney trials (Table 7 and Fig. 5). In Canberra, Sequence 1 developed 21% more fingermarks than Sequence 2. Whereas in Sydney, Sequence 1 developed 16% more fingermarks than Sequence 2.

Table 7
Results from the pseudo-operational trials on 5-year-old university examination booklets. (*Note*: To be counted, developed marks had to have a HO rating of 2 or above.).

	Sequence 1 (IND-Zn)		Sequence 2 (DFO)	
	Canberra	Sydney	Canberra	Sydney
No. pages/sides	40/80	10/20	40/80	10/20
No. marks with IND-Zn or DFO	332	87	248	61
No. marks with ninhydrin	67	19	73	29
No. marks with PD	5	4	12	5
Total marks for full sequence	404	110	333	95
Avg. marks per side: IND-Zn or DFO	4.2	4.4	3.1	3.1
Avg. marks per side: ninhydrin	0.8	1.0	0.9	1.5
Avg. marks per side: PD	0.1	0.2	0.2	0.3
Avg. marks per side: full sequence	5.1	5.5	4.2	4.8

4. Summary and conclusions

Extensive trials were undertaken in both Canberra and Sydney to evaluate the performance of two fingermark detection sequences on four common paper substrates and on 5-year-old university examination booklets. The two sequences considered started with either IND-Zn (Sequence 1) or DFO (Sequence 2), followed by ninhydrin, physical developer and nile red. Nile red was subsequently removed from consideration as it did not result in any fingermark development at the end of each sequence. Based on the donor trials, there was no significant difference in the performance observed for the two sequences overall. However, as a single method, IND-Zn clearly outperformed DFO. Ninhydrin performed better when used in sequence after DFO than when used after IND-Zn. In the pseudo-operational trials, the IND-Zn sequence outperformed the DFO sequence, indicated by a higher average number of fingermarks developed per treated page. While IND-Zn proved to be the single best development method of those tested, ninhydrin and PD, when used in sequence after IND-Zn, were still able to develop additional fingermarks. As a result, the following sequence is recommended for use on common paper substrates under Australian conditions:

IND-Zn → Ninhydrin → Physical Developer

Given the poor nile red results obtained in both Canberra and Sydney, the use of nile red at the end of a full sequence on paper substrates is of questionable value. While not verified, the lack of results with this reagent may be a result of the solvents and/or heat treatment used earlier in the sequence with the amino acid reagents. The use of nile red at the end of a full sequence is, therefore, not recommended. Nile red may still be useful in sequence after physical developer for samples that have previously been wet (i.e., where amino acid reagents are not employed); however, further experimental data is required to confirm this. Additional research is required to determine which fraction of the sebaceous deposit is being targeted by lipid stains such as nile red, compared to the components of the deposit being targeted by physical developer. The relative stability of these different fractions in the latent fingermark deposit also needs further study.

While a smaller number of samples were processed in the Sydney component of the study, the Canberra and Sydney trials produced similar results, with the same general trends being observed at both locations. This suggests that the different environmental conditions between the two sites did not play a significant role in the development of the latent fingermarks on the porous substrates tested. However, it should be noted that only a relatively small difference in average relative humidity (61% RH in Sydney compared to 50% RH in Canberra) was encountered over the period of the trials. Further research is recommended to extend the study to other locations with much wider differences in relative humidity.

Acknowledgements

The authors would like to thank Mackenzie de la Hunty, Centre for Forensic Science, University of Technology, Sydney, for her assistance in the early stages of this project.

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